Per-channel complex residuals to the same validation FRF

New fit begins at 3 Hz. Old-model behavior below 25 Hz is extrapolation.

Residuals are complex differences at identical original bins. The validation record is the second half of the same experiment.

Validation LRM magnitude — Old-model complex residual — Joint-model complex residual

$u_1 \to y_1$ $u_2 \to y_1$ $u_3 \to y_1$ $u_4 \to y_1$

$u_1 \to y_2$ $u_2 \to y_2$ $u_3 \to y_2$ $u_4 \to y_2$

$u_1 \to y_3$ $u_2 \to y_3$ $u_3 \to y_3$ $u_4 \to y_3$

$u_1 \to y_4$ $u_2 \to y_4$ $u_3 \to y_4$ $u_4 \to y_4$

Magnitude (dB, recorded output / motor-command unit)

Frequency (Hz)